



Jenkins Learning Roadmap (From Scratch)

◆ 1. What is Jenkins?

- Jenkins is an **open-source automation server**.
- It helps automate **build** → **test** → **deploy** processes.
- Mostly used for **Continuous Integration (CI)** and **Continuous Delivery (CD)** pipelines.

👉 Example:

You commit Selenium code → Jenkins pulls it → runs Maven → executes TestNG tests → generates reports → optionally deploys results.

◆ 2. Install Jenkins

Option A: Local installation (best for beginners)

- Install **Java 11/17** (Jenkins requires Java).
- Download Jenkins `.war` → run:
`java -jar jenkins.war`
- Open browser → `http://localhost:8080`

Option B: Docker installation (for practice with containers)

```
docker run -p 8080:8080 -p 50000:50000 jenkins/jenkins:lts
```

◆ 3. Basic Jenkins Setup

- Unlock Jenkins → create Admin user.
 - Install **Suggested Plugins** (Git, Maven, Email, Pipeline, TestNG).
 - Configure **JDK, Maven, Git** inside Jenkins (Manage Jenkins → Global Tool Configuration).
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◆ 4. Create First Job

👉 Go to **New Item** → **Freestyle Project**

Example: “Smoke Tests Job”

- **Source Code Management** → connect GitHub repo.
- **Build Tool** → choose Maven/Gradle.

- **Build Step** → `mvn clean test`.
- **Post-Build Action** → Publish TestNG or JUnit Reports.

Now every time you **click Build Now**, Jenkins will pull your repo, run tests, and show results.

◆ 5. Run Selenium/TestNG Framework in Jenkins

- Push your Java Selenium + TestNG project to GitHub.
 - Create a Jenkins job with:
 - `mvn clean test -DsuiteXmlFile=testng-smoke.xml`
 - Reports appear in Jenkins console + HTML reports if configured.
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◆ 6. Automate with Triggers

- **Poll SCM** → checks GitHub repo every X mins.
 - **GitHub Webhook** → triggers Jenkins build immediately after commit.
 - **Schedule Builds** (like cron jobs) → run daily/nightly regression at `0 0 * * *`.
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◆ 7. Pipeline (Jenkinsfile)

Instead of GUI jobs, you can write **Pipeline as Code** (Jenkinsfile).

Example:

```
pipeline {
    agent any
    stages {
        stage('Checkout') {
            steps {
                git 'https://github.com/harish/selenium-framework.git'
            }
        }
        stage('Build') {
            steps {
                sh 'mvn clean install'
            }
        }
        stage('Test') {
            steps {
                sh 'mvn test -DsuiteXmlFile=testng-regression.xml'
            }
        }
        stage('Report') {
            steps {
                publishHTML([reportDir: 'test-output', reportFiles:
'index.html', reportName: 'TestNG Report'])
            }
        }
    }
}
```

```
}  
    }  
}  
}
```

◆ 8. Reports & Notifications

- Integrate **Extent/Allure reports** → show fancy dashboards.
 - Add **Email/Slack notification plugin** → send pass/fail results to team.
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◆ 9. Advanced (After Basics)

- Distributed Builds → run Selenium on multiple nodes (Grid/Docker).
 - Parameterized Jobs → pass environment, browser, or test type (Smoke/Regression).
 - Integration with Docker, Kubernetes, AWS → CI/CD at scale.
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✓ Suggested 7-Day Plan for You

Day 1-2: Install Jenkins, create Freestyle jobs.
Day 3: Integrate with Git + Maven + TestNG project.
Day 4: Add triggers (SCM, schedule).
Day 5: Generate reports (Extent/Allure, HTML).
Day 6: Write your first Pipeline (`Jenkinsfile`).
Day 7: Add notifications + parameterized jobs.

👉 Harish, do you want me to prepare a **hands-on practice project for you** (like a Git repo with a simple Selenium TestNG framework + Jenkins integration), so you can directly practice without wasting time setting things up?